

Fiat Currency and Cryptocurrency: Two Sides of the Same Coin?

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August 10, 2026

Abstract

This study documents foreign exchange (FX) volatility as a positively priced cross-asset risk factor in the cross-section of cryptocurrency excess returns, implying that investors demand a premium for holding cryptocurrencies whose payoffs co-move with FX volatility under a standard risk-adjustment framework. To underscore the practical implications of this pricing power, we show that cross-asset time-series trading strategies conditioned on historical FX volatility are profitable. These findings align well with the currency competition theoretical framework and offer important insights for both academics and practitioners.

JEL Classification: G12, G15, F31

Keywords: cryptocurrency returns; FX volatility; asset pricing; trading strategy.

1 Introduction

Digital currencies, commonly known as cryptocurrencies, have recently garnered significant academic attention, spurring research into the cross-section of their expected returns (e.g., [Zhang et al., 2021](#); [Liu et al., 2022](#); [Borri and Shakhnov, 2022](#); [Fieberg et al., 2023](#); [Lee and Wang, 2024](#), among others). A common thread across these studies is their reliance on crypto-specific risk factors to explain the cross-sectional variation in cryptocurrency risk premia, consistent with standard asset-pricing paradigms in the [Fama and French \(1993\)](#) and [Carhart \(1997\)](#) models adapted to digital assets. A critical review of this small but growing literature reveals that, to our knowledge, no prior study has incorporated cross-market data—such as traditional risk factors from equities, bonds, or currencies—to price cryptocurrency premia, and the present study aims to address this gap.

Our primary research question investigates whether FX volatility risk is priced in cryptocurrency returns within a standard asset-pricing framework. We assess the cross-sectional implications of this question using a four-factor model based on FX volatility risk—serving as a proxy for global macroeconomic uncertainty ([Menkhoff et al., 2012a](#))—augmented by crypto market, size, and momentum factors, established in [Liu et al. \(2022\)](#) as robust proxies for systematic risk in cryptocurrency markets. The FX volatility risk factor, constructed from 10 major developed-market currencies, is tested against weekly excess returns for 30 major cryptocurrencies by market capitalization, spanning June 2, 2014 to October 28, 2024.

Our central finding is that FX volatility risk is priced in the cross-section of cryptocurrency returns, indicating that it captures information relevant for pricing not only fiat currencies ([Menkhoff et al., 2012a](#)) but also digital assets, implying common systematic factors underpin cross-sectional pricing across crypto and fiat currency markets. We find that cryptocurrency returns exhibit positive, significant FX volatility risk premia, suggesting that cryptocurrencies with higher exposure to FX volatility have significantly higher returns than cryptocurrencies with lower exposure to this factor. Within a standard risk–return framework, our evidence of a significantly positive risk premium on FX volatility risk (macro uncertainty proxy) implies that crypto investors are compensated for bearing exposure to this global FX-related risk.

We validate this novel evidence through several robustness tests across multiple specifications. The FX volatility factor retains significant pricing power when the test assets shift from individual cryptocurrencies to portfolios and when the analysis is conducted over subsample periods. FX volatility pricing persists even after controlling for established cryptocurrency return predictors (Caldara and Iacoviello, 2022; Liu et al., 2022; Dobrynskaya, 2024) and volatility spillovers from stock and bond markets. These findings reaffirm the critical robust role of FX volatility in capturing cryptocurrencies' exposures to common sources of risk.

Building on the established pricing power of the FX volatility factor in cross-sectional settings, we extend our analysis to time-series contexts and document a significant negative spillover relationship from FX volatility to cryptocurrency returns. We test the profitability implications of this result using a cross-asset trading strategy that employs moving-average (MA) technical trading signals, taking long (short) positions in a cryptocurrency when its short-term MA exceeds (falls below) its long-term MA. This strategy corroborates our asset pricing results and documents the economic success of our currency-based trading strategies, generating substantial statistical and economic profits across our cryptocurrency sample.

In a nutshell, FX volatility helps both price cryptocurrency returns and generate profits. Our time-series results show that FX volatility transmits systematic risk to crypto markets, while our cross-sectional results reveal that this monetary-competition risk is positively priced, so investors demand compensation for fiat-linked instability. Within a standard risk–return framework, the positive cross-sectional premium suggests that investors are compensated for bearing systematic FX-related risk, whereas the negative time-series response captures the realization of that risk. Together, both results portray cryptocurrencies as imperfect monetary substitutes that inherit systemic fiat risks and are compensated for bearing them.

We situate this new evidence within the theoretical framework of currency competition (Schilling and Uhlig, 2019; Benigno et al., 2022; Chiu and Koepl, 2022), where cryptos and fiat currencies imperfectly substitute as stores of value, and fiat monetary uncertainty—proxied by FX volatility—emerges as a fundamental pricing factor. Our findings can also be interpreted through several economic mechanisms, including the intertemporal hedging channel implied

by Merton's (1973) ICAPM ([Merton, 1973](#)), the Campbell–Shiller discount-rate channel ([Campbell and Shiller, 1988](#)), and the flight-to-safety or liquidity channel ([Anadu et al., 2025](#)).

From an academic perspective, we distinguish ourselves from the cryptocurrency literature by making the first attempt to explain the cross-section of cryptocurrency excess returns using cross-asset data from the FX market. Previous research focuses on explaining cross-sectional expected returns using cryptocurrency-specific factors, with an incomplete list of examples including market, size, and momentum ([Liu et al., 2022](#)), downside risk ([Zhang et al., 2021](#); [Dobrynskaya, 2024](#)), realized volatility ([Lee and Wang, 2024](#)), liquidity ([Borri and Shakhnov, 2022](#)), uncertainty ([Lucey et al., 2022](#)), technical indicators ([Fieberg et al., 2023](#)), and network factors ([Liu and Tsyvinski, 2021](#)), among others. We acknowledge that there have been few attempts to relate cryptocurrencies to currencies within the framework of currency competition ([Benigno et al., 2022](#)), causality ([Palazzi et al., 2021](#)), and risk management ([Ferroni, 2022](#)), but none has done so in an asset pricing framework, to our knowledge.¹

From a practical perspective, our work offers practitioners key insights for cryptocurrency trading by demonstrating that incorporating historical FX volatility information improves cryptocurrency position management. We regard this latter evidence as being complementary to prior research on cross-asset trading strategies that combine signals from equity and bond markets ([Pitkääjärvi et al., 2020](#)), equity, bond, and FX markets ([Phylaktis and Yamani, 2025](#); [Goulding and Harvey, 2025](#)), carry trade and commodities ([Lu and Jacobsen, 2016](#)), as well as crude oil volatility and stock markets ([Fernandez-Perez et al., 2023](#)).

The remainder of the paper proceeds as follows. Section 2 builds on the currency competition literature to present a stylized model linking FX volatility to cryptocurrencies. Section 3 documents the ability of FX volatility to price cryptocurrency returns, and Section 4 presents robustness tests. Section 5 documents the success of currency-based trading strategies in generating profits in the cryptocurrency market; Section 6 concludes.

¹Although several studies document the inter-relatedness of currencies and other asset classes such as stocks ([Dobrynskaya, 2014](#)), bonds ([Ballensiefen and Ranaldo, 2023](#)) as well as commodities and options ([Koijen et al., 2018](#)), the role of FX market information has surprisingly yet to be extended to cryptocurrencies.

2 Theoretical Framework

2.1 Motivation

The choice to use FX volatility risk to price cryptocurrency returns is inspired by an emerging literature that examines the connectedness between currencies and cryptocurrencies. From a theoretical viewpoint, the Crypto-Enforced Monetary Policy Synchronization (CEMPS) theory of [Benigno et al. \(2022\)](#) offers two channels for the currency-cryptocurrency relationship through the impact of currency competition between currencies and cryptocurrencies on interest rates and exchange rates (the first leg) as well as monetary policies (the second leg).

The first mechanism is driven by the classical literature on substitution theory and currency competition among fiat currencies ([Girton and Roper, 1981](#), [Kareken and Wallace, 1981](#), and [Matsuyama et al., 1993](#)) which has paved the way for a modern strand of literature that extends the analysis of currency competition to cryptocurrencies ([Schilling and Uhlig, 2019](#), [Fernández-Villaverde and Sanches, 2019](#), [Benigno et al., 2022](#), [Chiu and Koepl, 2022](#)). This line of research posits that private cryptocurrencies, unlinked from any sovereign entity, compete with local sovereign currencies, given that both currencies (physical and digital) are substitutes in providing liquidity services, thereby challenging national currency sovereignty.

[Benigno et al. \(2022\)](#), who examine currency substitution in a two-country model with complete markets featuring two national currencies and a global cryptocurrency that is used in both local markets, propose the CEMPS framework as a general stochastic structure through which cryptocurrencies affect interest rates and exchange rates. According to the CEPMS theory, the simultaneous circulation of global and local currencies requires the synchronization of monetary policies across countries, leading to equal national nominal interest rates.

To characterize the implications of the CEMPS for exchange rate dynamics, [Benigno et al. \(2022\)](#) argue that the two local currencies compete indirectly through the global currency, which enjoys worldwide acceptance and is used in both local markets, making the exchange rate between the national currencies a risk-adjusted martingale and inducing substantial exchange rate volatility. Along the same lines, [Schilling and Uhlig \(2019\)](#) analyze currency

competition between the US Dollar and the Bitcoin in an endowment economy, and show that Bitcoin, despite its price volatility, can be viewed as an international medium of exchange.

The second mechanism examines the implications of cryptocurrency competition for the international monetary system (Fernández-Villaverde and Sanches, 2019, Benigno, 2023, Benigno et al., 2022). It suggests that such competition constrains the central bank's ability to implement the monetary policy, including adjustments to benchmark interest rates and the money supply, thereby exerting significant effects on capital flows in cryptocurrency market.² Fernández-Villaverde and Sanches (2019), who build a model of currency competition among privately issued fiduciary currencies based on the framework of Lagos and Wright (2003), find that currency competition can lead to an inefficient social allocation of the money supply.

In this setup, cryptocurrencies limit central banks by constraining the effectiveness of their primary operational instrument – the interest rate (Benigno et al., 2022) and by restricting their ability to achieve core monetary policy objectives such as price stability and inflation control (Benigno, 2023). Under a currency substitution framework, central banks thus risk being trapped at the zero lower bound if they seek to lower the domestic interest rate in an attempt to preserve the role of the national currency as a medium of exchange by restricting the circulation of the global currency within the domestic economy.

2.2 Stylized Model

We capitalize on the currency competition theory (Von Hayek, 1976; Kareken and Wallace, 1981; Schilling and Uhlig, 2019) to present a model that links the stochastic discount factor (SDF) for cryptocurrency returns to FX volatility in an asset pricing framework. Consider a representative agent who allocates wealth W_t across two asset classes: (i) a fiat portfolio of sovereign currency bonds indexed by j , with gross nominal return $R_{j,t+1}$ and FX volatility exposure $\sigma_{FX,t}$; and (ii) a set of N cryptocurrencies indexed by i , serving as imperfect monetary substitutes for fiat money with returns $R_{i,t+1}$ and FX volatility exposure β_i^{FX} .

²For example, an increase in the U.S. interest rates may induce investors to reallocate capital from high-risk digital assets toward comparatively safer financial assets, such as U.S. government securities. Along the same lines, when the US dollar strengthens, cryptocurrency prices often fall, as the dollar's appreciation reduces the purchasing power of other currencies used to buy cryptocurrencies.

The agent maximizes lifetime utility from a composite consumption basket C_t denominated in U.S. dollars, following the standard [Cochrane \(2005\)](#) asset-pricing setting,

$$\max_{\{C_t, \omega_{i,t}\}} E_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\gamma}}{1-\gamma} \right], \quad (1)$$

where $\beta \in (0, 1)$ represents the subjective discount factor and $\gamma > 1$ the coefficient of relative risk aversion. As in [Schilling and Uhlig \(2019\)](#), the agent faces a cash-in-advance (CIA) constraint that requires a fraction of consumption to be financed by monetary assets,

$$P_t C_t \leq M_t^{\text{fiat}} + \phi \sum_{i=1}^N q_{i,t} X_{i,t}, \quad (2)$$

where $q_{i,t}$ denotes the price of cryptocurrency i , $X_{i,t}$ its quantity held, and ϕ the substitutability parameter ([Kareken and Wallace, 1981](#)) governing crypto i 's substitution for fiat currency j . Cryptocurrencies with a stronger monetary-substitution role become more valuable to hold precisely when FX volatility rises, since they help relax the CIA constraint when fiat currency markets become less reliable as a medium of exchange.

The agent's first-order condition for any asset i yields the standard no-arbitrage restriction,

$$1 = E_t[M_{t+1} R_{i,t+1}], \quad (3)$$

where M_{t+1} denotes the SDF derived from the agent's first-order condition,

$$M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \cdot \Psi_{t+1}, \quad (4)$$

and Ψ_{t+1} is the shadow price of the CIA constraint in Equation (2). A rise in FX volatility increases the value of cryptocurrencies' monetary-substitution service, which relaxes the CIA constraint and *lowers* its shadow price Ψ_{t+1} . By log-linearizing Equation (4) around the deterministic steady state ([Campbell and Shiller, 1988](#)) and substituting this channel, we get

$$m_{t+1} \equiv \log M_{t+1} \approx -\gamma \Delta c_{t+1} - \theta \Delta \sigma_{FX,t+1}, \quad (5)$$

where $\theta > 0$ denotes a reduced-form parameter increasing in ϕ , γ , and $\bar{\sigma}^2$, capturing the sensitivity of the CIA shadow price to FX volatility.

Equation (5) implies that the SDF loads *negatively* on FX volatility ($-\theta \Delta\sigma_{FX,t+1}$): an increase in FX volatility *lowers* the pricing kernel M_{t+1} , since it relaxes the agent's cash-in-advance constraint through cryptocurrencies' monetary-substitution channel rather than tightening it. High- $\Delta\sigma_{FX}$ states are thus *low*- M_{t+1} states — states in which the marginal value the agent places on an additional unit of payoff is relatively low. From the standard asset pricing relation (Cochrane, 2005), the expected excess return on cryptocurrency i is

$$E[r_i^e] = -\frac{\text{Cov}(m_{t+1}, r_{i,t+1}^e)}{E[M_{t+1}]} . \quad (6)$$

Substituting the SDF (5) into Equation (6), we get

$$E[r_i^e] = \underbrace{\gamma \text{Cov}(\Delta c, r_i^e)}_{\text{consumption risk premium}} + \underbrace{\theta \text{Cov}(\Delta\sigma_{FX}, r_i^e)}_{\text{FX volatility risk premium}} . \quad (7)$$

In currency competition equilibrium, the indifference condition (Schilling and Uhlig, 2019) requires cryptocurrency i 's expected return to equal fiat currency j 's return adjusted for the monetary risk premium,

$$\underbrace{\frac{q_{i,t+1} - q_{i,t}}{q_{i,t}}}_{\text{crypto } i \text{ capital gain}} = \underbrace{r_{j,t}}_{\text{fiat } j \text{ return}} + \underbrace{\lambda \beta_i^{FX}}_{\text{monetary risk premium}} - \underbrace{\phi \mu_{i,t}}_{\text{monetary service yield}} . \quad (8)$$

Equation (8) implies that crypto i must earn a higher expected capital gain than fiat j by the monetary risk premium $\lambda \beta_i^{FX}$, net of its monetary service yield $\phi \mu_{i,t}$, where λ is derived explicitly in Equation (10) below and β_i^{FX} is the cryptocurrency i 's FX-volatility beta,

$$\beta_i^{FX} \equiv \frac{\text{Cov}(\Delta\sigma_{FX,t+1}, r_{i,t+1}^e)}{\text{Var}(\Delta\sigma_{FX,t+1})} , \quad (9)$$

so that $\text{Cov}(\Delta\sigma_{FX}, r_i^e) = \beta_i^{FX} \text{Var}(\Delta\sigma_{FX})$, with no sign restriction imposed a priori: $\beta_i^{FX} > 0$ for cryptocurrencies whose returns rise when FX volatility rises, and $\beta_i^{FX} < 0$ for those that

fall. Substituting into Equation (7), the cross-sectional pricing equation becomes

$$E[r_i^e] = \alpha + \underbrace{\theta \cdot \text{Var}(\Delta\sigma_{FX})}_{\lambda > 0} \cdot \beta_i^{FX}. \quad (10)$$

Equation (10) constitutes the central result of the model: since $\theta > 0$ and $\text{Var}(\Delta\sigma_{FX}) > 0$, the price of FX-volatility risk $\lambda \equiv \theta \cdot \text{Var}(\Delta\sigma_{FX})$ is unambiguously *positive*, so cryptocurrencies with higher (more positive) FX-volatility exposure β_i^{FX} earn higher average excess returns. The economic mechanism is a hedging argument, not a level effect on any single asset's price: a positive- β_i^{FX} cryptocurrency delivers its best payoffs precisely in high- $\Delta\sigma_{FX}$ states, which Equation (5) identifies as *low*- M_{t+1} states — states in which the agent's marginal utility of an extra dollar is low. Such an asset offers no insurance against the risk the agent actually cares about and must therefore compensate investors with a higher expected return. Conversely, a negative- β_i^{FX} cryptocurrency pays off in high- M_{t+1} states, acting as a hedge, and so commands a lower (or negative) risk premium. On average across the cryptocurrency universe, whether higher FX volatility raises or lowers contemporaneous crypto prices therefore depends on the sign of the asset's own β_i^{FX} ; it is the cross-sectional spread in β_i^{FX} , not a uniform time-series effect, that Equation (10) prices.

This theoretical framework provides intuition for our empirical analysis and serves as a stylized motivational device, rather than a formal structural derivation of the empirical model developed in the next section. The log-linearization in Equation (5), the reduced-form treatment of θ , and the equilibrium indifference condition in Equation (8) are simplifications that illustrate *why* FX volatility exposure might plausibly carry a positive price of risk in the cryptocurrency cross-section, not a tightly identified model from which the empirical analysis in the next section is formally derived. Hence, the empirical specification in the next section should accordingly be read as motivated by, rather than nested within, this theoretical sketch.

3 Does FX Volatility Price Cryptocurrency Risk?

3.1 Cryptocurrency Returns

Our dataset comprises cryptocurrency data sourced from CoinCodex <https://coincodex.com>, alongside currency data obtained from DataStream. All variables are observed daily and aggregated to the weekly frequency using Monday-dated weekly averages.³ For notational clarity, cryptocurrencies are denoted by i , currencies by j , daily data by τ , and weekly data by t . Our sample begins on the earliest date for which data are available for at least five cryptocurrencies and spans June 2, 2014, to October 28, 2024, yielding 544 weekly observations. This approach accommodates differences in data availability across cryptocurrencies.

We analyze a sample comprising 30 major cryptocurrencies by market capitalization. We apply a series of data filters to the initial sample, which starts with all cryptocurrencies having a market capitalization exceeding 1 billion dollars. We then filter out coins lacking complete data on price, volume, and market capitalization; those with fewer than 3 years of data (i.e., issued after October 2021); and Tether (USDT) due to its status as a stablecoin. Finally, we filter extreme outliers in cryptocurrency returns by winsorizing at the 0.5% and 99.5% percentiles, effectively removing the top and bottom 1% of observations.

Table 1 reports the mean and standard deviation of weekly cryptocurrency returns (in percent), dollar trading volume (in billions), and market capitalization (in trillions) for the 30 cryptocurrencies over the full sample period. Daily returns are first calculated as logarithmic price changes for each cryptocurrency i on day τ , and weekly returns, denoted by $R_{i,t}$, are then computed as the average of daily returns over the seven trading days in week t . Results are also reported for an equally weighted portfolio of the 30 cryptocurrencies in our sample, which has an average weekly return of 0.094% and a standard deviation of 1.536%.

[Table 1 here.]

³We construct weekly series by averaging daily observations within each Monday–Sunday calendar week and dating each value to the Monday of that week. Crypto averages use seven daily observations, while FX averages use up to five trading days, so the two series share a common Monday-dated weekly index despite differing numbers of underlying daily observations. This procedure aligns the cryptocurrency and FX series on a common Monday-dated weekly index despite their differing numbers of daily observations.

3.2 FX Volatility Risk Factor

The FX volatility risk factor is constructed from information on WMR/Reuters spot exchange rates for the 10 major developed market currencies: Australian dollar (AUD), British pound (GBP), Canadian dollar (CAD), Danish krone (DKK), Euro (EUR), Japanese yen (JPY), New Zealand dollar (NZD), Norwegian krone (NOK), Swedish krona (SEK), and Swiss franc (CHF). All spot exchange rates are quoted from the point of view of a U.S. investor as the number of foreign currency units per one U.S. dollar.

We first construct a daily measure of global FX volatility, denoted as $fxvol_\tau$, by averaging the absolute daily log return for each currency j on day τ across all sample currencies available on that day. Our global FX volatility proxy in week t , denoted as $FXVOL_t$, is then given by average-of-the-week value of $fxvol_\tau$ across the 5 trading days per week, as follows

$$FXVOL_t = \frac{1}{T_t} \sum_{\tau \in T_t} fxvol_\tau = \frac{1}{T_t} \sum_{\tau \in T_t} \left[\sum_{j=1}^N \frac{|R_{j,\tau}|}{N_\tau} \right] \quad (11)$$

where $R_{j,\tau}$ denotes the log return of currency j on day τ , calculated as the logarithmic change in spot exchange rates, and N_τ represents the number of available currencies on day τ . This measure is similar in spirit to the FX volatility measure proposed by [Menkhoff et al. \(2012a\)](#).

To take a first look at the relation between FX volatility and cryptocurrency returns, the final two columns of Table 1 report the pairwise Pearson correlation coefficients (with associated p-values) between the two variables, thus assessing their average correlation over the entire sample period. The results indicate that FX volatility is significantly negatively correlated with the returns of all 30 cryptocurrencies in the sample, with correlation coefficients ranging from -0.028 (for PAX Gold) to -0.167 (for Algorand), and the cryptocurrency portfolio.

3.3 FX Volatility and the Cross-Section of Cryptocurrency Returns

Our goal is to examine whether FX volatility, proxied by Equation 11, is priced in cryptocurrency excess returns—defined as cryptocurrency returns less the risk-free rate, proxied by the one-month Treasury bill rate from the Federal Reserve Economic Data

(FRED) database—which serve as our test assets.⁴ To account for systematic risk in the cryptocurrency market, we construct cryptocurrency market, size, and momentum risk factors following the three-factor model of Liu et al. (2022), our benchmark for cryptocurrency return cross-sections. The cryptocurrency market excess return (CMKT) is defined as the difference between the cryptocurrency market return—calculated as the equal-weighted return of all available constituent coins—and the risk-free rate. For the cryptocurrency size factor (CSIZE), each week we partition the 30 coins in our sample into five size groups based on market capitalization; the CSIZE factor is then measured as the return differential between the small- and large-cap portfolios. The cryptocurrency momentum factor (CMOM) is constructed analogously by sorting cryptocurrencies into five groups based on their lagged returns, with CMOM defined as the return difference between the high- and low-momentum portfolios.

We test the cross-sectional implications of our hypothesis regarding the pricing power of FXVOL for cryptocurrency returns by estimating a linear four-factor model that includes FXVOL, CMKT, CSIZE, and CMOM, utilizing the standard two-pass regression methodology of Fama and MacBeth (1973) (FMB). In the first pass, we estimate time-series regressions of cryptocurrency excess returns $r_{i,t}$ for each cryptocurrency i in week t on the risk factors,

$$r_{i,t+1} = \alpha_i + \beta_{i,1}FXVOL_{t+1} + \beta_{i,2}CMKT_{t+1} + \beta_{i,3}CSIZE_{t+1} + \beta_{i,4}CMOM_{t+1} + \xi_{i,t+1}. \quad (12)$$

In the second pass, we run a cross sectional regression of average excess returns on the estimated betas from the first stage to estimate the prices of risk,

$$\bar{r}_{i,t} = \hat{\beta}_{1i}\lambda_{FXVOL} + \hat{\beta}_{2i}\lambda_{CMKT} + \hat{\beta}_{3i}\lambda_{CSIZE} + \hat{\beta}_{4i}\lambda_{CMOM} + \xi_t. \quad (13)$$

Subsequently, we refer to this four-factor model (as defined in Equations 12 and 13) as our baseline model that relies on the notion that cryptocurrency excess returns are exposed to systematic risk in the FX market (proxied by the FXVOL factor that extracts the global component in the FX market) that is additional to the information in systematic risk in the

⁴The merit of using a crypto-level analysis is to mitigate concerns regarding data-snooping biases or the use of too few portfolios as test assets (Lo and MacKinlay, 1990 and Lewellen et al., 2010).

cryptocurrency market (proxied by CMKT, CSIZE, and CMOM factors in [Liu et al., 2022](#)).

Table 2 presents the asset pricing results from the second pass of the FMB procedure (i.e., the estimated cross-sectional lambdas λ s in Equation 13, along with their associated [Newey and West \(1987\)](#) standard errors), where the test assets consist of the excess returns on the 30 cryptocurrencies in our sample. To assess the contribution of the FXVOL factor relative to established cryptocurrency risk factors in explaining cryptocurrency returns, we estimate three specifications of our baseline model that differ in their control sets. The first includes only FXVOL; the second adds CMKT and CSIZE; and the third further adds CMOM.

We are primarily interested in the estimated price of FXVOL risk. When FXVOL is specified as the sole factor, its estimated price of risk is positive and statistically significant at the 1% level, amounting to 4.9% per week. The estimated price of FXVOL risk is 0.4% per week after controlling for CMKT and CSIZE, and 0.9% per week when CMOM is also included. Hence, FX volatility is a priced risk factor in cryptocurrency markets, even after controlling for the benchmark cryptocurrency return determinants. This empirical evidence is in line with our theoretical prediction in Equation (10), which implies that FX volatility commands a positive price of risk in the cross-section of cryptocurrency returns.

We also report the cross-sectional model fit statistics – the [Hansen and Jagannathan \(1997\)](#) (HJ) for the null hypothesis that pricing errors are zero, the root mean squared pricing errors (RMSE), and the R-squared statistics. Our baseline model yields a high R-squared of about 88.8% and we cannot reject the null that the pricing errors are zero at the 10% level.

The last two rows of Table 2 report pairwise Pearson correlation coefficients between FXVOL factor and cryptocurrency risk factors (CMKT, CSIZE, and CMOM). The results indicate that FX volatility is negatively correlated with CMKT, with a correlation coefficient of -0.109 , but exhibits no significant correlation with CSIZE or CMOM.

[Table 2 here.]

4 Robustness Tests

In this section, we explore several robustness checks on our evidence regarding the power of FX volatility risk in pricing cryptocurrency returns, as shown in Table 2. To this end, we assess various specifications of our baseline asset pricing model using alternative test assets and sample periods. We also examine whether the pricing power of the FXVOL factor is driven by systematic risk in the cryptocurrency market or other financial markets.

4.1 Portfolio-level Analysis

As an alternative assessment of the role of FX-volatility risk in cryptocurrency returns, we conduct portfolio-level asset-pricing tests in Table 3 and compare the results with those for individual cryptocurrencies in Table 2, using two prominent performance criteria from the asset-pricing literature. In Panel A of Table 3, the test assets consist of equally weighted average excess returns on three sets of cryptocurrency quintile portfolios: (1) return portfolios sorted by lagged returns, (2) volume portfolios sorted by lagged trading volume, and (3) size portfolios sorted by lagged market capitalization.⁵ Overall, the FXVOL factor retains its pricing power when cryptocurrency portfolios serve as test assets, with results that are robust across alternative portfolio-sorting variables. Its estimated price of risk ranges from 0.7% per week for size portfolios to 4.4% for return portfolios and 5.6% for volume portfolios, indicating that FX volatility robustly prices cryptocurrency excess returns.

For completeness, Panel B of Table 3 repeats the same analysis in Panel A of Table 3 with the only exception that our test assets now comprise the value-weighted (in lieu of equally weighted) excess returns on cryptocurrency portfolios.⁶ Examining the overall results in

⁵For robustness, we also run our analysis using volatility portfolios (sorted on lagged realized volatility) and liquidity portfolios (sorted on lagged liquidity). We construct a daily realized volatility measure, defined as $cvol_{i,\tau} = |R_{i,\tau}|$, where $R_{i,\tau}$ denotes the log return of cryptocurrency i on day τ . We construct an analogous daily liquidity measure for each cryptocurrency, defined as $cliq_{i,\tau} = \frac{|R_{i,\tau}|}{V_{i,\tau}}$, where $V_{i,\tau}$ denotes the trading volume of cryptocurrency i on day τ . The weekly measures are then given by average-of-the-week values of $cvol_{i,\tau}$ and $cliq_{i,\tau}$. The results are similar to those reported in Table 3, so they are surpassed to save space.

⁶We first compute the daily market-capitalization weight of each cryptocurrency i on day τ as its market capitalization divided by the aggregate market capitalization of all cryptocurrencies in the sample, and then construct the daily value-weighted return as the weighted average of individual cryptocurrency returns. The weekly value-weighted returns are measured as the average of the daily returns for the seven days.

Panel B of Table 3, we find that the FXVOL factor continues to be positive and statistically significant at 1% level across the three variants of cryptocurrency portfolios, indicating that our findings remain robust across alternative cryptocurrency return measurement methods. This evidence underscores the pivotal role of the FXVOL factor in capturing cryptocurrencies' exposures to common sources of risk at both the individual-crypto and portfolio levels.

[Table 3 here.]

4.2 Subsample Analysis

We further investigate whether time-varying risk and return dynamics, driven by evolving market conditions, affect the pricing power of the FXVOL factor in cryptocurrency returns. To this end, Panel A of Table 4 presents the asset pricing results for our baseline model, estimated over three subsample periods centered around the COVID-19 period: (1) Pre-COVID period: June 2, 2014, to January 31, 2020 (Panel A); (2) Post-COVID period: February 1, 2020, to October 31, 2024 (Panel B); and (3) Bear market period: November 15, 2021, to October 31, 2024 (Panel D). The test assets consist of the excess returns on the 30 cryptocurrencies in our sample. Our subsample analysis in Panel A of Table 4 corroborates the full-sample evidence reported in Table 2, showing that FXVOL continues to have a significant and robust pricing effect on cryptocurrency excess returns across all three subsample periods.

For robustness, Panel B of Table 4 reports the asset pricing results using subsamples of varying sizes: specifically, the 20, 10, and 5 largest cryptocurrencies ranked by market capitalization. These three subsamples exclude Wrapped Bitcoin, a wrapped token, as well as Tether Gold and PAX Gold, which are tokenized gold assets rather than conventional cryptocurrencies. Once again, the FXVOL factor continues to be positive and statistically significant at 1% level across the three variants of samples, suggesting that the FXVOL factor's pricing power is not attributable to any particular subset of cryptocurrencies.

[Table 4 here.]

4.3 Controlling for Global Risk Factors

Our baseline model in Equation 13 uses the market, size, and momentum factors in Liu et al. (2022) to account for systematic risk in the cryptocurrency market. Nevertheless, the growing cryptocurrency literature suggests other cryptocurrency-specific risk factors.⁷ Therefore, our evidence on the FXVOL factor’s success in pricing cryptocurrency returns may be driven by omitted factors related to unexpected changes in cryptocurrency market conditions, such as price volatility, investor sentiment, and geopolitical risks.

To address this concern, we conduct a more stringent asset-pricing analysis by separately augmenting the baseline model in Equation 13 with each of the following three additional risk factors: (1) the downside market-risk factor (CMKTDR) of Zhang et al. (2021) and Dobrynskaya (2024), which captures adverse conditions in the cryptocurrency market; (2) the cryptocurrency music sentiment indicator (CMSI) of Hadhri et al. (2025), a proxy for aggregate investor mood that captures broad emotional states reflected in Spotify users’ music-listening choices; and (3) the geopolitical risk index (GPR) of Caldara and Iacoviello (2022), which quantifies risks associated with adverse geopolitical events.⁸

To assess the relative pricing power of the aforementioned three risk factors (CMKTDR, CMSI, and GPR) relative to the FXVOL factor, we conduct a horse-race test between these three control factors and the FXVOL factor, while controlling for CMKT, CSIZE, and CMOM in all model specifications. Table 5 reports the asset pricing results for this analysis. Finding that the inclusion of any of these control factors eliminates the statistical significance of the FXVOL factor would imply that its pricing power is driven by these factors. Examining the results in Table 5, we find no evidence that this is the case. The FXVOL factor remains significantly positive across all model specifications, while the estimated risk premia for each of the three control factors are also statistically significant, consistent with prior evidence (see Dobrynskaya, 2024; Hadhri et al., 2025; and Caldara and Iacoviello, 2022).

⁷See Borri et al. (2026) for a survey of empirical asset-pricing regularities in cryptocurrency markets.

⁸Following Dobrynskaya (2024), CMKTDR is defined as an interaction dummy variable ($\text{CMKTDR} = \text{CMKT} \times \text{Dummy}$) that takes the value of zero for negative CMKT values and one otherwise. We obtain the cryptocurrency music sentiment indicator (CMSI) data directly from the authors. GPR data are obtained from the website of Matteo Iacoviello: <https://www.matteoiacoviello.com/gpr.htm>.

So far, our overall results show that the FX volatility risk is priced in the cross section of cryptocurrency returns. This raises the question of whether volatility-based risk factors constructed from other financial markets can subsume the pricing power of FXVOL in explaining cryptocurrency returns. To address this question, Table 5 reports the results for two alternative five-factor specifications, each of which augments our baseline model with either (1) equity market volatility (EQVOL) or (2) sovereign bond market volatility (BOVOL).⁹ We first construct a daily measure of global equity volatility, denoted $eqvol_\tau$, by averaging the absolute daily log equity returns for each country j on each day τ across all countries on any given day. The weekly global equity market volatility proxy, $EQVOL_t$, is then given by average-of-the week value of $eqvol_\tau$ across the 5 trading days per week. The $BOVOL_t$ factor is constructed in a similar fashion, using quotes for sovereign bond indices. Table 5 also reports the response of cryptocurrency returns to the change in the VIX index and the term spread.¹⁰ Our goal is to assess whether the pricing power of the FXVOL factor can be explained by relying on its covariance with variables from equity and bond markets.

Overall, the evidence in Table 5 continues to underscore the important role of FX volatility in capturing cryptocurrencies' exposures to common sources of risk. Table 5 shows that FXVOL estimates remain consistently positive and statistically significant in the presence of equity and bond variables. Interestingly, even after controlling for the CMKT, CSIZE, and CMOM factors, both the EQVOL and BOVOL factors bear significantly risk prices of 3% and -5%, respectively, both significant at the 1% level. These findings suggest that volatility information from FX, equity, and bond markets helps in pricing cryptocurrency returns, indicating that cryptocurrencies are not fully insulated from traditional financial system shocks, with their risk premiums influenced by global foreign exchange stress.

[Table 5 here.]

⁹Data for stock and bond returns are obtained from Datastream. We use data for Datastream stock market indices and 5-year Datastream government bond indices for our sample of 10 developed countries (Australia, Canada, Germany, Japan, New Zealand, Norway, Switzerland, Sweden, Denmark and the UK).

¹⁰Data for the VIX index and the term spread are obtained from the FRED database website.

4.4 Systematic Risk Spillovers

To better elucidate the connection between cryptocurrencies and fiat currencies, we investigate the extent to which established return predictors in FX markets are reflected in cryptocurrency returns, and conversely, whether predictors from cryptocurrency markets are reflected in currency returns. To this end, Table 6 presents a comparative asset-pricing analysis of cryptocurrency returns (Panel A) and currency returns (Panel B) with respect to prominent determinants of returns in the FX and cryptocurrency markets, respectively.

On one hand, Panel A of Table 6 reports asset pricing results for an alternative specification of our baseline model, replacing FXVOL with three prominent currency-based risk factors: the dollar (DOL) factor, the carry (CAR) factor, and the currency momentum (MOM) factor, all of which have been shown to price currency excess returns in the FX market literature (Lustig et al., 2011 and Menkhoff et al., 2012b).¹¹ The test assets are the excess returns of the 30 cryptocurrencies in our sample. On the other hand, Panel B of Table 6 repeats the same analysis as in Panel A except that now we use the excess returns of the 10 developed market currencies in our sample (in lieu of cryptocurrency ones) as test assets and the DOL, CAR, and MOM factors (in lieu of the CMKT, CSIZE, and CMOM factors).

Two interesting findings emerge from the comparative analysis in Table 6, relating to both the statistical significance and the signs of the estimated risk prices. First, cryptocurrency returns are exposed to systematic risk originating in the FX market that is distinct from and additional to the information conveyed by systematic risk in the cryptocurrency market, and vice versa. Specifically, Panel A reveals that not only the FXVOL factor but also the DOL, CAR and MOM factors are priced in the cross-section of cryptocurrency returns, even in the presence of popular determinants of cryptocurrency returns. In comparison, Panel B shows that the estimated risk prices of CMKT, CSIZE, and CMOM are statistically significant at the 1% level when evaluated with respect to currency returns, even after controlling for systematic risk in the FX market (proxied by DOL, CAR, and MOM factors).

¹¹The dollar factor is defined as the average currency excess returns of five single-sorted forward discount portfolios. The carry factor is measured as the currency return differential between the portfolios with the highest and lowest forward discount. The currency momentum factor is calculated as the currency return differential between the portfolios with the highest and lowest lagged currency return.

Second, we capture asymmetric responses to currency-based factors across cryptocurrency and currency returns in several instances. Put differently, many of our estimated risk prices flip sign when switching from cryptocurrency returns to currency returns as test assets. For example, we find that cryptocurrency excess returns display a significantly negative sensitivity to both CAR and MOM factors in Panel A of Table 6. In contrast, currency excess returns display a consistently significant and positive pricing relation with both variables across almost all model specifications in Panel B of Table 6, a finding which is consistent with [Lustig et al. \(2011\)](#) and [Menkhoff et al. \(2012b\)](#). These sign flips are also observed for the FXVOL factor, which is our main variable of interest. While our overall results indicate that FXVOL commands a significantly positive cross-sectional price of risk with respect to cryptocurrency returns, its estimated risk price is proven to be significantly negative when currency portfolios are considered ([Menkhoff et al., 2012a](#)). These findings suggest that common systematic factors underpin cross-sectional pricing across cryptocurrency and fiat currency markets.

[Table 6 here.]

This established cross-market nexus, captured in Sections 3 and 4, operationalizes the stylized model outlined in Section 2 and can be understood within the theoretical framework of currency-competition between traditional fiat currencies and cryptocurrencies ([Schilling and Uhlig, 2019](#), [Benigno et al., 2022](#), [Chiu and Koepl, 2022](#)), particularly with respect to interest rates and exchange rates. Within this framework, cryptocurrencies, when used alongside national currencies, act as a disciplinary force on monetary policy, inducing pressure toward interest-rate synchronization and, in some cases, amplifying exchange-rate volatility and financial-stability risks ([Kroszner 2007](#)). The shared risk factors arising from currency-substitution pressures help explain the joint cross-sectional pricing dynamics between cryptocurrencies and sovereign currencies amid intensifying policy-convergence pressures.

5 Cross-Asset Trading Strategies: Time-Series View

In this section, we move beyond cross-sectional asset-pricing analysis and adopt a time-series, practical perspective to study whether FX volatility can be used as a trading signal to construct profitable cryptocurrency strategies. Despite its importance, the cryptocurrency literature offers no evidence on the economic value of currency-based signals. We take a step in that direction by developing cross-asset trading strategies to study how much crypto investors can gain from trading conditional on FX-market cross-asset signals. Our approach complements prior work on cross-asset trading strategies that combine information from stock and bond markets (Pitkääjärvi et al., 2020), carry trade and commodity markets (Lu and Jacobsen, 2016), and crude oil volatility and equity markets (Fernandez-Perez et al., 2023).

5.1 The Cross-Asset Spillover Effects of FX Volatility

To set the stage for developing cross-asset trading strategies, we first assess the significance and direction of time-series linkages in FXVOL spillovers between the cryptocurrency and FX markets. To this end, we use a vector autoregressive model with error terms following a GARCH-in-mean process, a model known as the VAR-GARCH-M (Engle and Kroner, 1995),

$$\begin{cases} Y_t = M \cdot FXVOL_t + A \cdot Factors_t + R \cdot Y_{t-1} + \varepsilon_t \\ H_t = Z' r_{t-1} r'_{t-1} Z + G' H_{t-1} H'_{t-1} G \end{cases} \quad (14)$$

where

$$Y_t = \begin{bmatrix} r_{p,t}^{Crypto} \\ r_{p,t}^{FX} \end{bmatrix}, \quad M = \begin{bmatrix} m_{11} \\ m_{21} \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix}, \quad R = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix}$$

Equation 14 allows us to analyze the simultaneous response of cryptocurrency returns ($r_{p,t}^{Crypto}$) and currency returns ($r_{p,t}^{FX}$)¹² to FXVOL spillovers through the M -matrix elements m_{11} and m_{21} , which measure the spillover to cryptocurrency and currency returns, respectively.

¹²The equally weighted cryptocurrency portfolio combines individual cryptocurrencies to yield a return calculated simply as average returns on the individual cryptocurrencies: $r_{p,t}^{Crypto} = \frac{1}{N} \sum_{i=1}^N (R_{i,t})$, where N is the set of available cryptocurrencies at week t . Analogously, the equally weighted currency portfolio return is defined as average returns on the individual 10 developed market currencies: $r_{p,t}^{FX} = \frac{1}{10} \sum_{j=1}^{10} (R_{j,t})$.

To control for potential cross-asset return linkages between the two markets, we include two additional interaction channels. The A -matrix captures the joint impact of cryptocurrency risk factors (CMKT, CSIZE, CMOM) on both portfolio returns, while the R -matrix captures cross-market return spillovers and autoregressive dynamics. The off-diagonal elements r_{12} and r_{21} of the R -matrix capture how cryptocurrency and currency returns respond to each other's lagged returns, while the diagonal elements r_{11} and r_{22} capture their responses to their own lags. The Z and G -matrices encode the GARCH parameters.

Table 7 reports the VAR-GARCH-M estimates for the elements of the M , A , and R matrices, using $r_{p,t}^{Crypto}$ (Panel A) and $r_{p,t}^{FX}$ (Panel B) as test assets. To assess the importance of cross-market effects in our analysis, we estimate three different specifications of Equation 14 that vary in the set of control variables included with the FXVOL factor. The first specification includes only the cryptocurrency risk factors (i.e., the elements of the A -matrix), the second includes the lagged crypto and currency returns (i.e., the elements of the R -matrix), and the third includes both. Consequently, this analysis consists of six regressions.

Our primary interest is to examine the significance and direction of FX volatility spillovers on cryptocurrency returns. In Panel A of Table 7, we find a significant negative impact of FX volatility on cryptocurrency returns across all model specifications, lending credence to our theoretical framework in Equation 8 which implies that a rise in FX volatility depresses cryptocurrency prices contemporaneously. This evidence is further supported by the pairwise Pearson correlations in Table 1, which show a negative correlation of -0.11 between the cryptocurrency portfolio return and FX volatility, significant at the 5% level.

A contrasting pattern emerges when we examine the response of currency returns to FX volatility in Panel B of Table 7. Interestingly, the relationship between the two variables turns out to be insignificant once cross-market effects are taken into account. The cross-market return spillover results suggest that there is a unidirectional positive return spillover from cryptocurrency markets to currency markets (given by significant estimates of r_{21}), suggesting that currency returns react positively to cryptocurrency returns in the previous period.

[Table 7 here.]

5.2 Cross-Asset Time-Series Moving Average

The single-asset time-series technical trading strategies have been widely used in the literature on stocks, bonds, currencies (Moskowitz et al., 2012), commodities (Georgopoulou and Wang, 2017), and futures (Kim et al., 2016), where investors rely on single-asset predictors derived from technical indicators to make investment decisions, betting on return continuation. Motivated by our evidence of a significant relationship between cryptocurrency returns and FX volatility, we extend prior research on single-asset cryptocurrency technical signals (Liu et al., 2022; Fieberg et al., 2023) and design a cross-asset trading strategy using past cryptocurrency returns (same-asset signal) and FX volatility (cross-asset signal) as joint predictors.

Within this framework, we design a cross-asset trading strategy within the Moving-Average framework. We begin by computing the weekly short- and long-horizon moving averages of FX volatility, $MA_t^{FXVOL} = \frac{1}{j} \sum_{w=0}^{j-1} FXVOL_{t-w}$ for $j = s, l$, where s (l) is the length of the short (long) MA ($s < l$). We then construct the weekly MA-based long-short trading signals, denoted by $Z_{i,t}^{TSMA}$, assuming that investors take a long position in cryptocurrency i when its short-horizon moving average, $MA_{s,t}$, is greater than or equal to its long-horizon moving average, $MA_{l,t}$; otherwise, they take a short position. The $Z_{i,t}^{TSMA}$ can be specified as follows

$$Z_{i,t}^{TSMA} = \begin{cases} 1, & \text{if } MA_{s,t}^{FXVOL} \geq MA_{l,t}^{FXVOL}, \\ -1, & \text{if } MA_{s,t}^{FXVOL} < MA_{l,t}^{FXVOL}. \end{cases} \quad (15)$$

Given such a setup, the excess return on our cross-asset time series MA strategy, denoted as $r_{i,t}^{TSMA}$, equals the actual crypto return multiplied by the trading signal, as follows

$$r_{i,t}^{TSMA} = \begin{cases} +1 \times r_{i,t}, & \text{if } r_{i,t-w} > 0 \text{ and } Z_{i,t}^{TSMA} = 1. \\ -1 \times r_{i,t}, & \text{if } r_{i,t-w} < 0 \text{ and } Z_{i,t}^{TSMA} = -1. \\ 0, & \text{otherwise.} \end{cases} \quad (16)$$

Table 8 presents the profitability evaluation results for this MA-based trading strategy, using three weekly performance measures – mean, volatility, and skewness. To conserve space, Panel A of Table 8 reports these statistics only for the five cryptocurrencies with the largest

market capitalizations – Bitcoin, Ethereum, XRP, Binance Coin, and Solana. Overall, we find that our cross-asset strategy performs well. To assess whether the profitability effect diminishes over longer horizons, we examine the profitability of four weekly moving-average indicators with short-window lengths of $s = 10$ and 20 weeks and long-window lengths of $l = 50$ and 100 weeks. We denote these indicators as MA(10,50), MA(10,100), MA(20,50), and MA(20,100), all of which yield consistent results and generate economically meaningful profits. For example, the MA(10,50) strategy generates positive and statistically significant returns of 4.72% for Bitcoin, compared with 4.88% for Ethereum and 4.61% for XRP.

Panel B of Table 8 reports the economic evaluation results for an equally weighted portfolio that combines individual cryptocurrencies to yield a return calculated simply as average returns on individual cryptocurrencies: $TSM A_t = \frac{1}{N} \sum_{i=1}^N (TSM A_{i,t})$, where N is the set of available cryptocurrencies at week t . Our evidence, on the profitability of the cross-asset MA strategy, is also supported for the cross-sectional cryptocurrency portfolio, which yields a high and statistically significant return of 4.68%. Similar results are obtained for the medium- and long-horizon MA strategies, which use moving-average windows of 20, 50, and 100 weeks.

[Table 8 here.]

A potential concern is that the results reported in Table 8 are driven primarily by the single-asset trading signal, $r_{i,t-w}$, in Equation 16. To address this concern, we construct an alternative specification of our MA-based strategy that follows the preceding approach but bases its trading signal solely on the cross-asset trading signal, $Z_{i,t}^{TSM A}$. The excess return on this single-predictor MA strategy, denoted by $r_{i,t}^{TSM A, robust}$, is given by the realized cryptocurrency return multiplied by the corresponding trading signal, $Z_{i,t}^{TSM A}$, as follows

$$r_{i,t}^{TSM A, robust} = \begin{cases} +1 \times r_{i,t}, & \text{if } Z_{i,t}^{TSM A} = 1. \\ -1 \times r_{i,t}, & \text{if } Z_{i,t}^{TSM A} = -1. \end{cases} \quad (17)$$

Table 9 reports the profitability evaluation of this single-predictor MA-based trading strategy. The empirical setup is identical to that in Table 8, except that the trading signal is based on a single predictor rather than joint predictors. Panel A of Table 9 shows that the

long-short MA payoffs for all the top five cryptocurrencies are significantly positive across the four MA trading rules. Similar results are observed for the cross-sectional portfolio in Panel B of Table 9, where MA(10,50) and MA(10,100) yield significant positive weekly returns of 2.12% and 2.60%, respectively (compared to 2.10% and 2.44% for the 20-week MA).

[Table 9 here.]

6 Conclusion

In this study, we document that FX volatility is an empirically powerful variable in explaining cryptocurrency returns from both asset-pricing and trading-strategy perspectives. From an asset-pricing standpoint, we show that FX volatility is positively priced in the cross-section of cryptocurrency returns (Table 2), suggesting that cryptocurrencies with greater exposure to FX volatility earn higher average returns, as investors demand a premium for holding assets whose payoffs co-move with FX volatility under a standard risk-adjustment framework. This pricing power remains robust when (1) cross-sectional portfolios, rather than individual cryptocurrencies, are used as test assets (Table 3); (2) subsample periods and sets of cryptocurrencies are considered (Table 4); and (3) standard determinants of cryptocurrency returns, as well as equity and bond volatilities, are included as controls (Table 5). We also document bidirectional systematic-risk spillovers: cryptocurrency risk factors affect currency markets, while currency risk factors, in turn, affect cryptocurrency markets (Table 6). This evidence highlights a two-way dependence in cross-sectional pricing and shared risk dynamics.

From a trading-strategy perspective, we find that cryptocurrency excess returns fall on average when FX volatility rises contemporaneously in a time-series context (Table 7). To highlight the practical implications of this result, we construct two hypothetical long-short cryptocurrency trading strategies: (1) a strategy that uses both past cryptocurrency returns (a same-asset signal) and FX volatility (a cross-asset signal) as joint predictors (Table 8); and (2) a strategy that relies solely on FX volatility as the predictor (Table 9). The results document the economic success of both cross-asset trading strategies in generating both high statistical significance and economically positive returns for our sample of cryptocurrencies.

In sum, we identify FX volatility, viewed as a proxy for global macroeconomic uncertainty, as a positively priced and statistically significant risk factor in cryptocurrency markets. Although increases in FX volatility are associated with lower cryptocurrency returns in the time series, cryptocurrencies with greater FX-volatility exposure earn significantly higher average returns in the cross section. Within a standard risk–return framework, this positive cross-sectional premium indicates that investors are compensated for bearing systematic FX-related risk, whereas the negative time-series response reflects the realization of that risk.

From a theoretical viewpoint, this cross-market nexus between traditional currencies and cryptocurrencies is framed well within the currency competition literature ([Schilling and Uhlig, 2019](#); [Benigno et al., 2022](#); [Chiu and Koepl, 2022](#)), which posits both as substitutes for liquidity services that foster competition undermining national monetary control. This framework forces monetary policy convergence but generates FX volatility and financial instability risks, suggesting that shared risk factors shape cross-sectional pricing of cryptocurrencies and currencies, highlighting their mutual sensitivity and interconnected risk dynamics. Our evidence—of a positive price of risk coupled with a negative contemporaneous impact—can be reconciled within several channels, including the intertemporal hedging channel ([Merton, 1973](#)), the Campbell–Shiller discount-rate decomposition channel ([Campbell and Shiller, 1988](#)), and the flight-to-safety or liquidity channel ([Anadu et al., 2025](#)).

This study should inspire further research in the cryptocurrency asset-pricing literature and in real-world cryptocurrency trading, as our findings indicate that FX volatility plays an important role in pricing cryptocurrency returns and in creating profitable trading opportunities. From an academic standpoint, our work should motivate future research that deepens the understanding of the drivers of the FX volatility factor and its relationship with cryptocurrency risk premia. From a practical perspective, our findings should encourage cryptocurrency traders to refine their position management by incorporating the informational content of historical FX volatility into their cryptocurrency return forecasts.

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Table 1: Descriptive Statistics

Cryptocurrency variable	Crypto symbol	$R_{i,t}$		$V_{i,t}$		$MV_{i,t}$		$FXVOL_t$	
		Mean	SD	Mean	SD	Mean	SD	ρ	Pvalue
Bitcoin	BTC	0.13	1.371	2215.425	306.668	2537.140	194.001	-0.088	0.041
Ethereum	ETH	0.232	2.054	2164.535	311.054	2423.119	237.476	-0.121	0.012
Ripple	XRP	0.116	2.262	1902.048	366.956	2244.214	210.306	-0.095	0.027
Binance	BNB	0.113	1.876	2040.230	168.054	2332.289	174.736	-0.106	0.049
Solana	SOL	0.375	2.465	2090.127	192.689	2284.006	221.576	-0.092	0.163
Tron	TRX	0.079	2.083	2011.322	145.357	2192.321	97.148	-0.110	0.034
Dogecoin	DOGE	0.148	2.383	1730.505	392.116	2020.789	270.273	-0.087	0.043
Cardano	ADA	0.066	2.144	1970.212	159.143	2269.695	129.796	-0.088	0.093
Bitcoin Cash	BCH	0.018	2.436	2044.669	95.115	2251.521	67.088	-0.130	0.012
Monero	XMR	0.115	2.23	1683.746	322.393	2003.881	253.220	-0.075	0.080
Chain Link	LINK	0.152	2.424	1912.343	246.531	2133.549	167.356	-0.136	0.009
UNUS SED LEO	LEO	0.081	0.865	1464.112	114.765	2167.048	61.606	-0.036	0.547
Stellar	XLM	0.073	2.299	1656.810	385.599	2022.473	255.939	-0.059	0.173
Zcash	ZEC	-0.073	2.355	1844.499	150.692	2009.973	107.557	-0.126	0.010
Avalanche	AVAX	0.119	2.671	2000.050	111.420	2239.902	126.226	-0.115	0.092
Litecoin	LTC	0.054	1.874	1917.019	290.392	2137.434	167.540	-0.089	0.039
Hedera	HBAR	0.001	2.007	1769.736	129.098	2046.516	147.581	-0.140	0.023
Cronos	CRO	0.001	2.06	1670.670	142.105	2122.925	120.900	-0.112	0.050
Polkadot	DOT	-0.018	1.934	2032.333	81.526	2292.043	69.270	-0.092	0.176
Uni Swap	UNI	0.004	2.113	1979.891	65.733	2223.720	78.885	-0.124	0.071
AAVE	AAVE	0.085	2.341	1957.419	69.821	2124.681	60.877	-0.122	0.077
OKB	OKB	0.1	2.016	1779.064	139.411	2125.498	134.681	-0.052	0.341
NEAR Protocol	NEAR	0.11	2.61	1957.512	98.306	2157.687	94.794	-0.163	0.018
Eth. Classic	ETC	0.068	2.262	1957.794	184.844	2107.841	110.629	-0.121	0.012
Tether Gold	XAUT	0.031	0.289	1376.323	193.904	1927.243	96.601	-0.039	0.545
PAX Gold	PAXG	0.021	0.29	1595.150	77.064	1961.342	68.240	-0.028	0.678
KuCoin Token	KCS	-0.034	2.119	1499.946	152.016	1957.102	113.743	-0.105	0.047
Algorand	ALGO	-0.089	2.075	1885.093	88.842	2081.049	130.101	-0.167	0.006
Filcoin	FIL	-0.167	2.211	1989.022	72.529	2166.849	61.891	-0.135	0.050
Wrapped Bitcoin	WBTC	0.146	1.268	1714.070	274.565	0.000	0.000	-0.158	0.006
Portfolio		0.094	1.536	1753.464	255.591	2043.905	149.042	-0.110	0.010

This table presents the summary statistics of cryptocurrency variables under study over the sample period from June 2, 2014 to October 28, 2024. The table reports the mean and standard deviation of weekly returns, $R_{i,t}$ (in percent), dollar trading volume, $V_{i,t}$ (in billions), and market capitalization, $MV_{i,t}$ (in trillions), for 30 major cryptocurrencies ranked by market capitalization. The row “Portfolio” reports statistics for an equally weighted cryptocurrency portfolio comprising all cryptocurrencies in the sample. The final two columns report pairwise Pearson correlation coefficients and associated p -values between the returns of individual cryptocurrencies—including the equally weighted cryptocurrency portfolio—and the FX volatility risk factor.

Table 2: FX Volatility Risk and the Cross-Section of Cryptocurrency Returns

		FXVOL	CMKT	CSIZE	CMOM	HJ	RMSE	R ² (%)
1	$\hat{\lambda}$	0.049***				38.923	0.028	82.0
	s.e.	0.002				0.069		
2	$\hat{\lambda}$	0.004**	-1.566***	0.020***		14.654	0.021	88.07
	s.e.	0.002	0.017	0.006		0.069		
3	$\hat{\lambda}$	0.009***	-1.576***	0.014**	0.057***	25.306	0.019	88.8
	s.e.	0.002	0.017	0.006	0.010	0.069		
Correlation Matrix								
		CMKT	CSIZE	CMOM				
ρ		-0.109	0.038	0.004				
Pvalue		0.011	0.382	0.925				

The table presents cryptocurrency-level cross-sectional results from a linear four-factor asset-pricing model that includes the FX volatility (FXVOL) risk factor and controls for the cryptocurrency market (CMKT), size (CSIZE), and momentum (CMOM) factors in [Liu et al. \(2022\)](#). The test assets are the excess returns of the 30 cryptocurrencies in our sample, as listed in Table 1. The table presents the second-stage results from the Fama–MacBeth asset-pricing tests: the estimated prices of risk (λ , as defined in Equation 13), their associated Newey–West standard errors, and cross-sectional statistics – the [Hansen and Jagannathan \(1997\)](#) (HJ) distance measure, the root mean squared pricing errors (RMSE), and the R-squared statistics. The final two rows report the pairwise Pearson correlation coefficients, ρ , and associated p -values between the FX volatility risk factor and the three cryptocurrency risk factors: CMKT, CSIZE, and CMOM. *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 3: FX Volatility Risk: Portfolio-Level Analysis

Panel A: Equally-Weighted Returns							
	FXVOL	CMKT	CSIZE	CMOM	HJ	RMSE	R ² (%)
Panel A.1: Return Portfolios							
$\hat{\lambda}$	0.044***	-1.333***	0.065***	0.026***	0.031	0.0030	96.4
s.e.	0.004	0.021	0.006	0.002	0.043		
Panel A.2: Volume Portfolios							
$\hat{\lambda}$	0.056***	-1.289***	-0.047***	-0.174***	0.618	0.0047	95.2
s.e.	0.002	0.015	0.002	0.007	0.043		
Panel A.3: Size Portfolios							
$\hat{\lambda}$	0.007***	-1.590***	0.004***	0.125***	0.168	0.0029	97.5
s.e.	0.002	0.016	0.001	0.006	0.043		
Panel B: Value-Weighted Returns							
	FXVOL	CMKT	CSIZE	CMOM	HJ	RMSE	R ² (%)
Panel B.1: Return Portfolios							
$\hat{\lambda}$	0.013***	-1.007***	-0.030***	0.001***	7.736	0.004	98.58
s.e.	0.001	0.001	0.000	0.000	0.043		
Panel B.2: Volume Portfolios							
$\hat{\lambda}$	0.276***	-0.974***	-0.010***	-0.079***	6.215	0.003	98.99
s.e.	0.002	0.001	0.000	0.001	0.043		
Panel B.3: Size Portfolios							
$\hat{\lambda}$	0.021***	-1.060***	-0.006***	0.132***	3.552	0.002	99.72
s.e.	0.001	0.001	0.000	0.001	0.043		

The table presents portfolio-level cross-sectional results from a linear four-factor asset-pricing model based on the FXVOL, CMKT, CSIZE, and CMOM factors. The test assets are the average excess returns of quintile cryptocurrency portfolios formed by sorting the 30 cryptocurrencies in our sample on prior-week excess returns, trading volume, and market capitalization. Panel A reports equally weighted portfolio returns, whereas Panel B reports value-weighted portfolio returns. In each panel, all cryptocurrencies are sorted into five portfolios based on their sorting variable, so that one-fifth of cryptocurrencies with the lowest (highest) values of the sorting variable are allocated to the first (fifth) portfolio. The table presents the price of risk estimates (lambdas, λ , in Equation 13) along with their associated Newey–West standard errors from the second-stage of the FMB asset-pricing tests and the cross-sectional statistics – the HJ distance measure, RMSE, and R-squared. *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 4: FX Volatility Risk: Subsample Analysis

Panel A: Subsample Periods							
	FXVOL	CMKT	CSIZE	CMOM	HJ	RMSE	R ² (%)
Panel A.1: Pre-COVID (June 2, 2014 - January 31, 2020)							
$\hat{\lambda}$	0.004***	-1.029***	0.035***	0.079***	10.120	0.004	93.37
s.e.	0.001	0.017	0.010	0.014	0.058		
Panel A.2: Post-COVID (February 1, 2020 - October 31, 2024)							
$\hat{\lambda}$	0.015***	-1.673***	-0.028***	0.055***	14.286	0.025	84.65
s.e.	0.000	0.025	0.001	0.003	0.069		
Panel A.3: Bear Market (November 15, 2021 - October 31, 2024)							
$\hat{\lambda}$	0.013***	-1.826***	-0.017***	0.007***	6.308	0.027	83.07
s.e.	0.000	0.027	0.001	0.002	0.079		
Panel B: Subsample Cryptocurrencies							
	FXVOL	CMKT	CSIZE	CMOM	HJ	RMSE	R ² (%)
Panel B.1: 20 cryptocurrencies							
$\hat{\lambda}$	0.007***	-1.560***	0.018***	0.055***	7.697	0.016	91.97
s.e.	0.002	0.017	0.006	0.010	0.068		
Panel B.2: 10 cryptocurrencies							
$\hat{\lambda}$	0.113***	-1.400***	0.189***	0.403***	6.861	0.009	96.45
s.e.	0.015	0.041	0.022	0.043	0.066		
Panel B.3: 5 cryptocurrencies							
$\hat{\lambda}$	0.022***	-1.656***	-0.046***	0.198***	2.958	0.002	99.69
s.e.	0.004	0.030	0.010	0.011	0.066		

The table presents subsample results for the linear four-factor asset-pricing model comprising FXVOL, CMKT, CSIZE, and CMOM. Panel A presents the results across three subsample periods: (1) Pre-COVID period (June 2, 2014 - January 31, 2020), (2) Post-COVID period (February 1, 2020 - October 31, 2024), and (3) Bear market period (November 15, 2021 - October 31, 2024). The test assets are the excess returns of the 30 cryptocurrencies in our sample. Panel B reports the results across three subsample cryptocurrencies: the 20, 10, and 5 largest cryptocurrencies by market capitalization. Each panel reports the estimated risk prices (along with Newey–West standard errors) in addition to the cross-sectional statistics (the HJ distance measure, RMSE, and R-squared). *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 5: FX Volatility Risk: Controlling for Financial and Macroeconomic Variables

	stat.	FXVOL	CMKT	CSIZE	CMOM	CMKTDR	CMSI	GPR	EQVOL	BOVOL	VIX	TERM
1 $\hat{\lambda}$		0.010***	-1.486***	0.029***	0.089***	-0.237***						
s.e.		0.002	0.026	0.008	0.014	0.013						
2 $\hat{\lambda}$		0.007***	-1.537***	-0.016***	0.041***		-0.114***					
s.e.		0.001	0.023	0.002	0.004		0.013					
3 $\hat{\lambda}$		0.011***	-1.628***	0.016**	0.075***			1.976***				
s.e.		0.003	0.019	0.007	0.012			0.405				
4 $\hat{\lambda}$		0.013***	-1.507***	0.016**	0.087***				0.030***			
s.e.		0.003	0.025	0.007	0.011				0.005			
5 $\hat{\lambda}$		0.006***	-1.668***	-0.034***	0.048***					-0.050***		
s.e.		0.002	0.029	0.011	0.009					0.012		
6 $\hat{\lambda}$		0.012***	-1.563***	0.017**	0.062***						0.029	
s.e.		0.004	0.021	0.007	0.012						0.023	
7 $\hat{\lambda}$		0.010***	-1.375***	-0.033***	-0.054**							9.149***
s.e.		0.002	0.044	0.009	0.021							1.757

The table reports asset-pricing results—including the estimated prices of risk (λ), Newey–West standard errors, and cross-sectional statistics—for augmented versions of the linear four-factor model comprising FXVOL, CMKT, CSIZE, and CMOM. Each specification separately controls for one of seven additional variables: (1) the cryptocurrency downside market-risk factor (CMKTDR) of [Dobrynskaya \(2024\)](#); (2) the cryptocurrency music sentiment indicator (CMSI) of [Hadhri et al. \(2025\)](#); (3) the geopolitical risk index (GPR) of [Caldara and Iacoviello \(2022\)](#); (4) equity market volatility (EQVOL); (5) bond market volatility (BOVOL); (6) changes in the VIX; and (7) the term spread. Both the EQVOL and BOVOL are constructed using our sample of 10 developed countries: Australia, Canada, Germany, Japan, New Zealand, Norway, Switzerland, Sweden, Denmark, and the United Kingdom. The test assets are the average excess returns of individual cryptocurrencies. *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 6: Cryptocurrency and Currency Asset Pricing under Shared Risks

Panel A: Test Assets: 30 Cryptocurrencies						
statistic	CMKT	Csize	CMOM	DOL	CAR	MOM
1 $\hat{\lambda}$	-1.664***	0.020**	0.083***	-0.015***		
s.e.	0.036	0.009	0.023	0.004		
2 $\hat{\lambda}$	-1.5751***	-0.0158***	0.007		-0.012***	
s.e.	0.017	0.003	0.006		0.001	
3 $\hat{\lambda}$	-1.557***	-0.016***	0.035***			-0.025***
s.e.	0.017	0.003	0.006			0.002
Panel B: Test Assets: 10 Developed Market Currencies						
statistic	DOL	CAR	MOM	CMKT	Csize	CMOM
1 $\hat{\lambda}$	-0.001***	0.003***	0.001***	-0.418***		
s.e.	0.000	0.000	0.000	0.114		
2 $\hat{\lambda}$	-0.001***	0.003***	0.001**		0.049***	
s.e.	0.000	0.000	0.000		0.012	
3 $\hat{\lambda}$	-0.001***	0.003***	0.001			0.021***
s.e.	0.000	0.000	0.000			0.007

The Table compares the asset pricing results across cryptocurrency returns (Panel A) and currency returns (Panel B) as test assets. Panel A reports results for three different specifications of a four-factor model, which differ in the choice of risk factors from the set DOL, CAR, and MOM, while controlling for CMKT, Csize, and CMOM factors in all model specifications. Panel B presents results for three alternative specifications of a four-factor model, where each specification varies the risk factor set among CMKT, Csize, and CMOM, while controlling for DOL, CAR, and MOM factors in all model specifications. In each panel, we report the price of risk estimates along with Newey–West standard errors from the second-stage of the FMB asset-pricing tests. *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 7: The Cross-Asset Spillover Effects of FX Volatility: Time-Series Evidence

Panel A: Cryptocurrency Returns						
statistic	FXVOL _t	CMKT _t	CSIZE _t	CMOM _t	$r_{i,t-1}^{crypto}$	$r_{j,t-1}^{FX}$
1 coeff.	-4.432***	0.043***	0.350***	0.324***		
s.e.	0.339	0.002	0.026	0.025		
2 coeff.	-3.508***				0.765***	0.231
s.e.	0.441				0.030	0.319
3 coeff.	-2.275***	0.043***	0.194***	0.185***	0.439***	-0.286
s.e.	0.301	0.000	0.023	0.022	0.026	0.221
Panel B: Currency Returns						
statistic	FXVOL _t	CMKT _t	CSIZE _t	CMOM _t	$r_{i,t-1}^{crypto}$	$r_{j,t-1}^{FX}$
1 coeff.	-0.009	0.001***	0.011***	-0.000		
s.e.	0.053	0.000	0.004	0.004		
2 coeff.	0.126**				0.020***	-0.008
s.e.	0.055				0.004	0.048
3 coeff.	0.069	0.001***	0.005	-0.005	0.016***	0.002
s.e.	0.058	0.000	0.004	0.004	0.005	0.042

The table presents the time-series coefficient estimates (and the associated standard errors) of the M, A, and R-matrices from the mean equation of the VAR GARCH-M model in Equation 14. The M-matrix captures the joint spillover of FX volatility to cryptocurrency returns and currency returns; the A-matrix allows the two market returns to be simultaneously influenced by cryptocurrency risk factors (CMKT, CSIZE, and CMOM); and the R-matrix controls for cross-market return spillovers from one market to the other, as well as responses to each market's own lagged returns. The test assets are cryptocurrency (Panel A) and currency (Panel B) portfolio returns, each assigning equal weights to all cryptocurrencies and developed-market currencies, respectively. *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 8: Cross-Asset Time-Series Moving-Average

Panel A: Top five cryptocurrencies				
	MA(10,50)	MA(10,100)	MA(20,50)	MA(20,100)
Bitcoin				
Mean	4.72***	5.03***	4.79***	4.94***
St. Dev.	0.03	0.03	0.03	0.03
Skewness	-0.44	-0.56	-0.46	-0.52
Ethereum				
Mean	4.88***	4.97***	4.93***	4.90***
St. Dev.	0.03	0.04	0.04	0.04
Skewness	-0.33	-0.35	-0.34	-0.31
XRP				
Mean	4.61***	4.90***	4.67***	4.81***
St. Dev.	0.04	0.04	0.04	0.04
Skewness	-0.19	-0.34	-0.22	-0.29
Binance				
Mean	4.57***	3.85***	4.55***	3.66***
St. Dev.	0.04	0.04	0.04	0.04
Skewness	-0.26	0.15	-0.23	0.25
Solana				
Mean	4.01***	3.92***	4.11***	3.86***
St. Dev.	0.04	0.04	0.04	0.04
Skewness	0.05	0.04	0.01	0.07
Panel B: Equally-Weighted Portfolio (XTSMOM)				
	MA(10,50)	MA(10,100)	MA(20,50)	MA(20,100)
Mean	4.68***	4.96***	4.74***	4.86***
St. Dev.	0.03	0.03	0.03	0.03
Skewness	-0.38	-0.46	-0.40	-0.41

The table presents the performance of the cross-asset time-series moving-average strategy that uses cryptocurrency returns (same-asset signal) and FX volatility (cross-asset signal) as joint predictors. The strategy takes a long (short) position in a cryptocurrency when its past weekly return is positive (negative) and the MA trend-following signal, defined in Equation 15, equals +1 (−1); otherwise, it takes no position. The table reports the mean return, volatility, and skewness of the trading strategy, whose return is defined in Equation (16), for the five largest cryptocurrencies by market capitalization (Panel A) and an equally weighted cryptocurrency portfolio (Panel B). Results use short MAs of 10 and 20 weeks and long MAs of 50 and 100 weeks. *, **, *** denote statistical significance 10%, 5% and 1% levels, respectively.

Table 9: Cross-Asset Time-Series Moving-Average: Single Predictor

Panel A: Top five cryptocurrencies				
	MA(10,50)	MA(10,100)	MA(20,50)	MA(20,100)
Bitcoin				
Mean	2.47***	3.10***	2.62***	2.90***
St. Dev.	0.07	0.06	0.07	0.07
Skewness	-0.75	-0.92	-0.78	-0.85
Ethereum				
Mean	2.43***	3.06***	2.37***	2.97***
St. Dev.	0.07	0.07	0.07	0.07
Skewness	-0.73	-0.85	-0.69	-0.82
XRP				
Mean	2.39***	3.03***	2.54***	2.84***
St. Dev.	0.07	0.07	0.07	0.07
Skewness	-0.62	-0.81	-0.66	-0.74
Binance				
Mean	2.37***	1.98***	2.02***	1.40*
St. Dev.	0.07	0.07	0.07	0.07
Skewness	-0.74	-0.59	-0.63	-0.42
Solana				
Mean	1.03	1.30	1.17	2.26***
St. Dev.	0.07	0.07	0.07	0.07
Skewness	-0.37	-0.47	-0.42	-0.67
Panel B: Equally-Weighted Portfolio (XTSMOM)				
	MA(10,50)	MA(10,100)	MA(20,50)	MA(20,100)
Mean	2.12***	2.60***	2.10***	2.44***
St. Dev.	0.06	0.06	0.06	0.06
Skewness	-0.71	-0.84	-0.72	-0.78

The table reports the performance of the cross-asset time-series moving-average strategy that uses FX volatility as its sole predictive signal. The strategy takes a long (short) position in a cryptocurrency when the moving-average trend-following indicator defined in Equation 15 equals +1 (−1). The table reports the mean return, volatility, and skewness of the trading strategy, whose return is defined in Equation (17), for the five largest cryptocurrencies by market capitalization (Panel A) and an equally weighted cryptocurrency portfolio (Panel B). Results are reported for short MA windows of 10 and 20 weeks and long MA windows of 50 and 100 weeks. *, **, *** denote statistical significance 10%, 5% and 1% levels, respectively.